A CARD GAA-family recurrence

Left: Neuron Right: Oligodendrocyte

	≥1		≥2		≥3		≥5	
Pure GAA repeat	29	26	8	2	4	0	2	0
Pure GAAA repeat	92	115	17	12	8	4	2	1
Pure GGAA repeat	72	93	18	23	8	10	3	5
Other 4 bp GAA	3	7	0	0	0	0	0	0
Other 5-10 bp GAA	63	80	15	17	8	4	3	0

Minimum donors with an expansion

B Cross-cohort replication of GAA-element Neuron loci

Source cohort \ Replication cohort	CARD (155 donors)	SMaHT (13 donors)	4-brain benchmark
CARD (155 donors)	—	11/132 8.3%	8/132 6.1%
SMaHT (13 donors)	11/21 52.4%	—	11/21 52.4%
4-brain benchmark	8/25 32.0%	11/25 44.0%	—

C GAA-element Neuron loci found in all three cohorts

N/O = donors with Neuron/Oligodendrocyte expansion

	chr10:48654648-48655296 (CTTT)	FGF14 chr13:102161539-102161869 (GAA)	chr19:33757354-33757634 (AAAG)	chr1:51906786-51906926 (AAAGAAAGG)	SH3RF3 chr2:109199301-109199876 (AATGG)	chr9:28189170-28189834 (AGGA)
4-brain benchmark	N 2 / O 0	N 2 / O 0	N 1 / O 0	N 1 / O 1	N 1 / O 0	N 1 / O 0
SMaHT (13 donors)	N 3 / O 2	N 1 / O 1	N 2 / O 2	N 1 / O 1	N 1 / O 1	N 4 / O 1
CARD (155 donors)	N 6 / O 1	N 1 / O 0	N 1 / O 0	N 3 / O 4	N 17 / O 0	N 12 / O 4